



Well, that was quick!

EVGA has already announced the availability of its GTX Titan X HydroCopper edition GPU primed for liquid cooling http://dgit.in/EVGA_TitanX

Lego Jurassic Park

A daughter along with her father recreated the original Jurassic Park using Lego's worth over \$1,00,000 <http://dgit.in/LegoJur>

What does Robotics Engineering involve?

Robotics Engineering deals with the development, design, operation and application of robots run by computer systems for their control, sensory feedback and information processing. Most of the robotics systems present today are being developed on the principles of electronic engineering, mechanical engineering and computer science and mathematics. So, the Robotics Engineer is a professional who has an in-depth knowledge about mechanical engineering with specific skills and expertise in the field of robotics.



Aakash Sinha, Founder and CEO, Omnipresent Robot Technologies Pvt. Ltd.

Fellow-IEEE and faculty at The Robotics Institute, Carnegie Mellon University said, "The growth in robotics industry in great, but it is not always recognised as "robotics". There is no strong distinction between automation and robotics. Demand is high, but many employers do not have good understanding of what graduates of robotics programs can do for them."

"There will be a huge demand for engineers in the field of robotics in the US and other developed countries", says Pankaj Saxena, Retail IT Consultant and Founder of Brampton Robotics, and also believes that India can utilise its creative talent in defences and other industries in the country.

What's the demand like for Robotics Engineers?

India is slowly becoming a hub for the production of industrial robots. Many American, Korean and Japanese firms are considering setting up Indian manufacturing centers. Jayant Keswani, Principal Partner & CEO at Milagrow Business and Knowledge Solutions states, "Robotics as an industry has seen significant growth in the past year and Milagrow expects the momentum to only increase in the next few years. Since there isn't any defined government policy on robotics currently, the category has a few small companies and startups which are charting its growth path."

Abheek Bose of Robot Alive, a Bengaluru-based robot technology startup, adds, "We should expect robotics and automation to be extensively adopted in industry and service sectors in another 5-10 years. Once that is established, there will be significant demand for robotics personnel and experts in the market."

There are many opportunities for robotics engineers both, in India and abroad. A professional with an M.E. degree in Robotics can get jobs in industries like space research, banking, gaming, medicine and intelligence and police departments.

Speaking about the world wide growth in the robotics sector, Mel Siegel, PhD,

What are educational requirements for Robotics Engineer?

Robotics Engineers are required to have at least a bachelor's degrees for entry-level jobs. The degree holder in mechanical engineering, related engineering specialties, physical science or mathematics could qualify for a job in robotics engineers. In addition to the educational qualification, you also need to have good problem solving skills, especially concerning mechatronics and other robot-related problems. To solve technical issues, you'll first understand the issue, simplify it and then apply your creativity to innovatively solve problems. Since the robotics discipline is mainly mathematics and physics-intensive, it would be wise to emphasise on these particular subjects at the school level. "Currently, it's difficult to find people with robotics skills. We end up training all our recruits. More than skills, we look for passion in the candidate and robotics involves multiple skills in areas such as

mechanical engineering, electronics and computer science," added Aakash, the finalist of US Defence's DARPA Robotics Grand Challenge. In other words, you should only choose this career if you're completely fascinated by robotics!

Which institutions offer courses in robotics?

Robotics is the field of scientific study that familiarises you with the design and creation of robots. More specifically, as a robotics student, you'll learn how to program a robot and understand its various applications. Some educational institutes offer undergraduate degrees in robotics engineering, which teach students how to apply math and mechanics to create automated machines. In India, Robotics can be studied at the IITs in Mumbai, Chennai, Delhi, Kanpur and Kharagpur. Other institutions that offer similar courses are the Indian Institute of Science at Bengaluru, National Institute of Technology, University of Hyderabad, Jadavpur University, Sri Sathya Sai Institute of Higher Learning (M.Tech) in Prasanthi Nilayam in Tamil Nadu, PSG College of Technology (M.Tech with Robotics as an elective subject) in Coimbatore in Tamil Nadu and Birla Institute of Technology and Science.

Institutes such as the Indian Institute of Robotics at Meerut in UP cater to those seeking a niche and highly specialised qualification in this sphere.

To earn a Ph.D. in Robotics, students must also generally submit a proposed plan for study and a dissertation idea related to areas such as artificial intelligence, robotics and kinetics, or robotics and sensors.

A few organisations such as LearnSocial and

Mumbai-based Techno Gravity Solution offer online courses and programs that expose students to the subject. You can also look online for tutorials on websites such as PlayWithRobots to guide you while building robots. Remember to look for programs approved by the Accredita-



Mel Siegel, PhD and faculty at the Robotics Institute, Carnegie Mellon University